DataAutoPilot Analysis Report

Generated at: 2026-03-14 04:04:25

CLASSIFICATION | Target: Survived | Shape: (891, 12)

Dataset:

Best Model

LogisticRegression


```
{
  "cv_f1": 0.8277911457535628,
  "f1": 0.7058823529411765,
  "accuracy": 0.8044692737430168,
  "tuning_top2": [
    {
      "model": "RandomForestClassifier",
      "tuned": true,
      "best_score": 0.7512937073929405,
      "best_params": {
        "model__n_estimators": 200,
        "model__min_samples_split": 10,
        "model__min_samples_leaf": 1,
        "model__max_features": "log2",
        "model__max_depth": null
      },
      "test_f1": 0.7142857142857143,
      "test_acc": 0.7988826815642458
    },
    {
      "model": "SVM",
      "tuned": false,
      "reason": "No tuning grid for svc",
      "test_f1": 0.7575757575757576,
      "test_acc": 0.8212290502793296
    }
  ]
}
```


Leaderboard

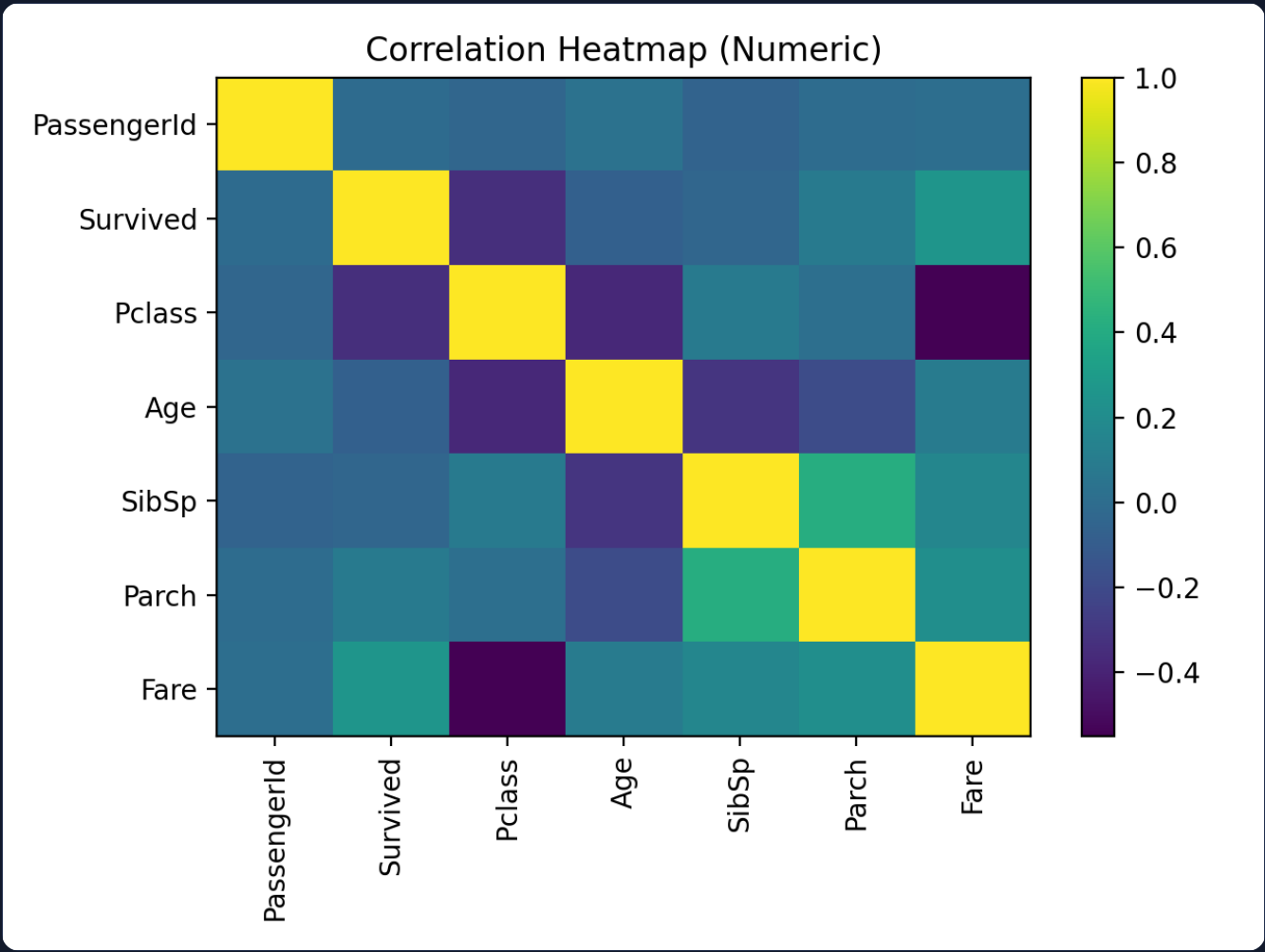
Model	Acc	Prec	Recall	F1	CV_F1
RandomForestClassifier	0.8044692737430168	0.84	0.6086956521739131		0.8277911457535628
SVM	0.8212290502793296	0.7936507936507936	0.7246376811594203		0.821260250778089
KNNClassifier	0.7988826815642458	0.7796610169491526	0.6666666666666666		0.8118162334151127
LogisticRegression	0.8268156424581006	0.8064516129032258	0.7246376811594203		0.805771741702849
XGBClassifier	0.7486033519553073	0.6818181818181818	0.6521739130434783		0.8037279155761278
DecisionTree	0.770949720670391	0.7121212121212122	0.6811594202898551		0.7964235014180463

Notes

- **note**: Leaderboard + top2 tuning + SHAP (best model only)

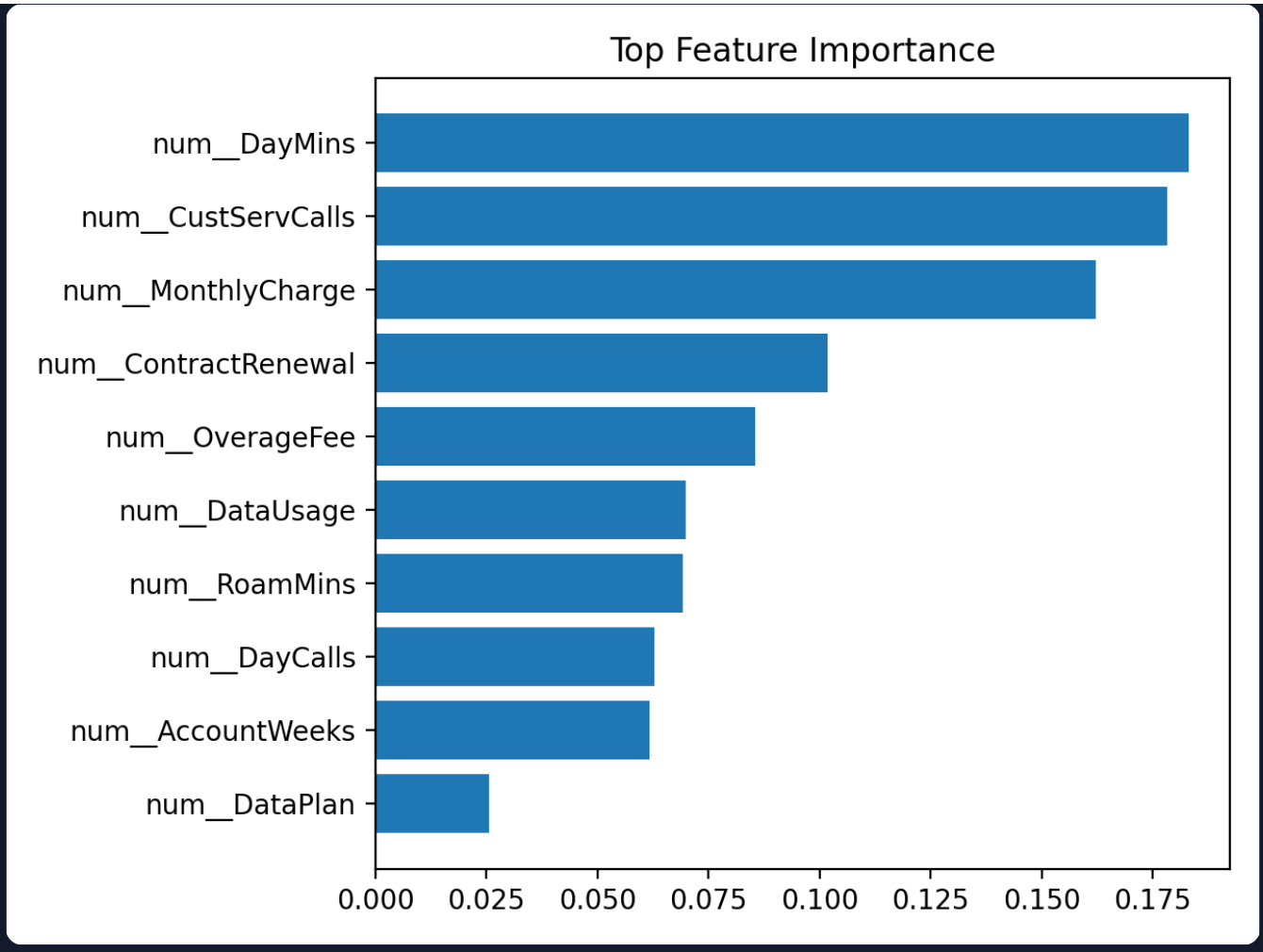
Artifacts

Correlation Heatmap



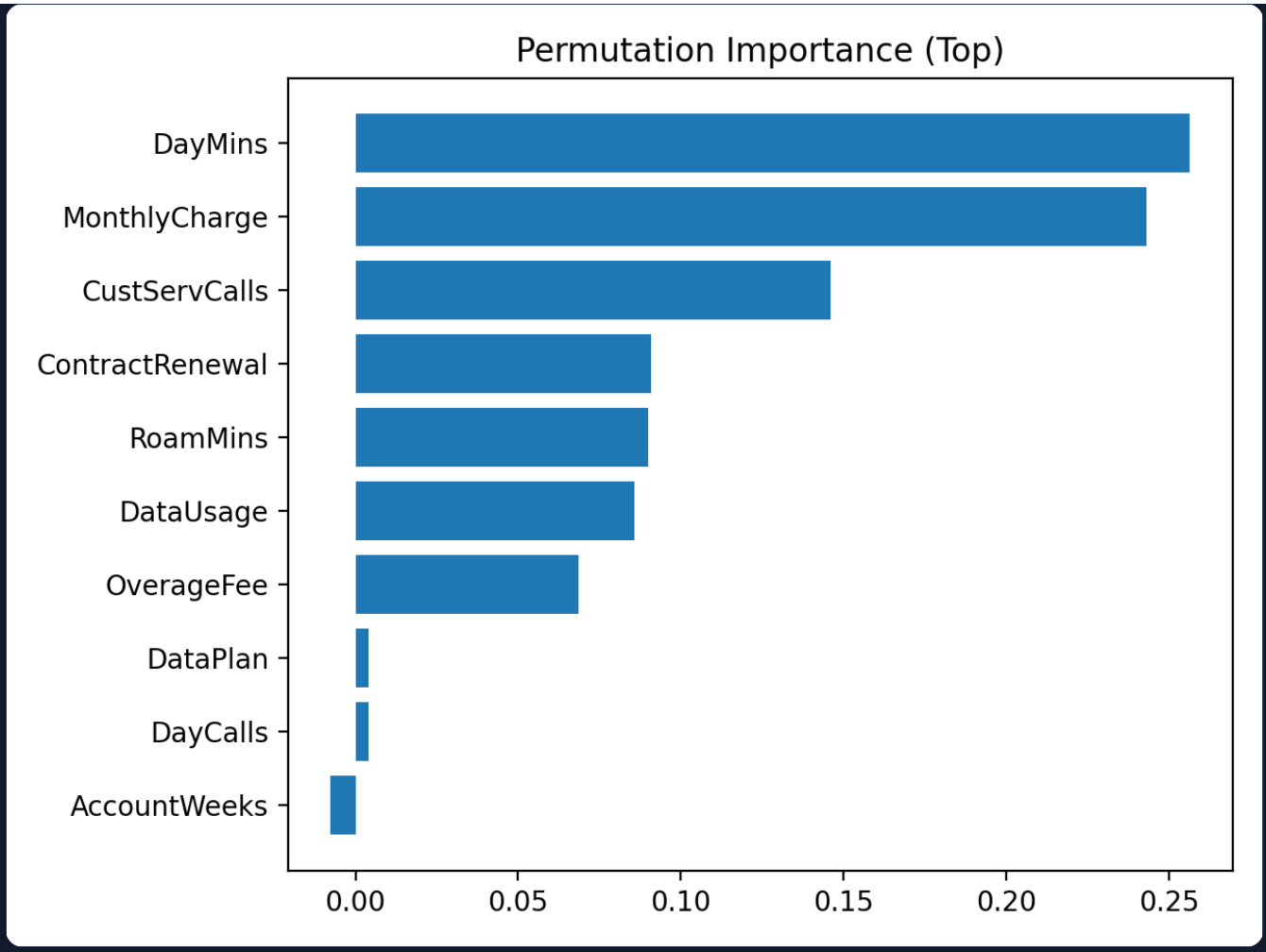
eda/correlation_heatmap.png

Feature Importance



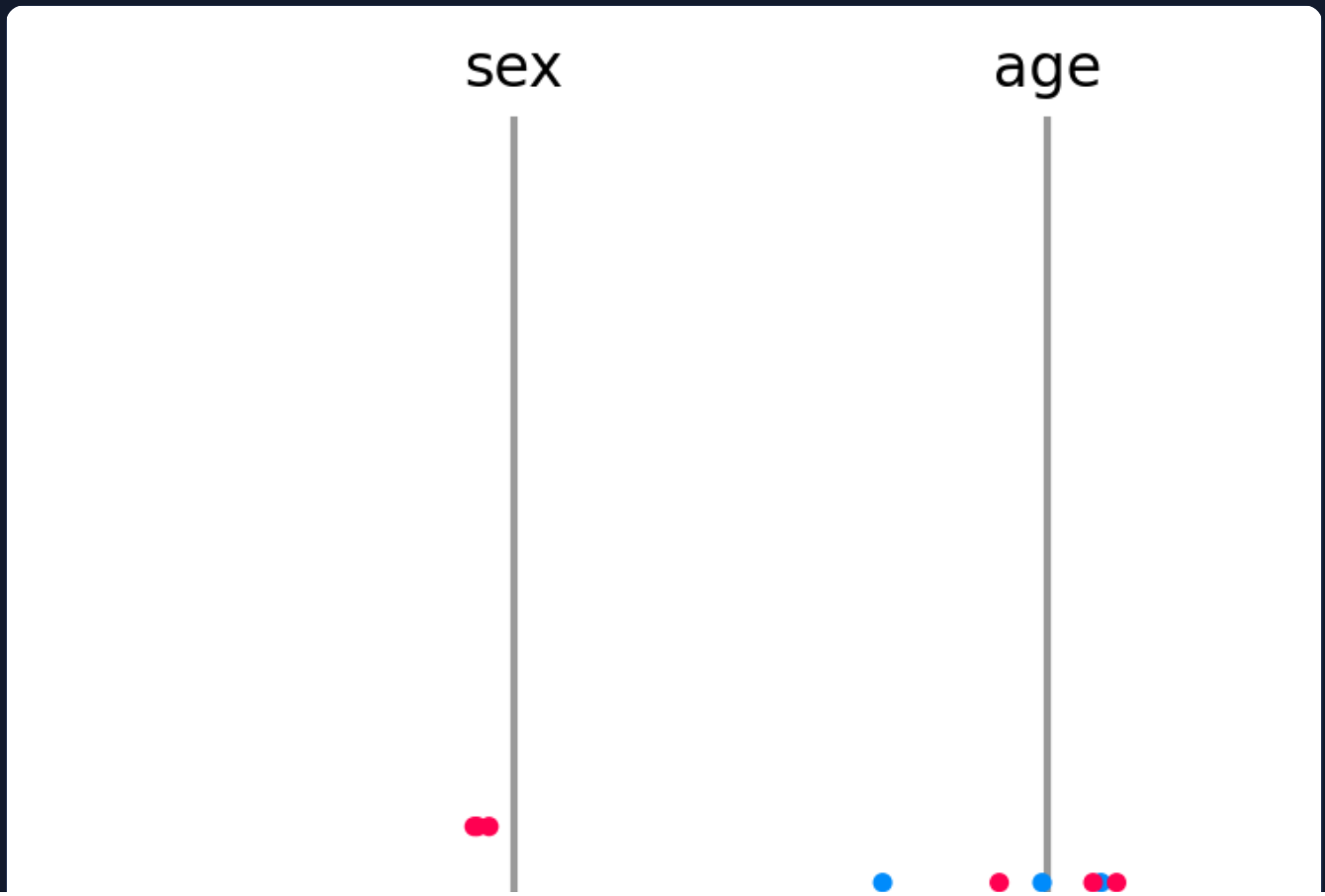
feature_importance.png

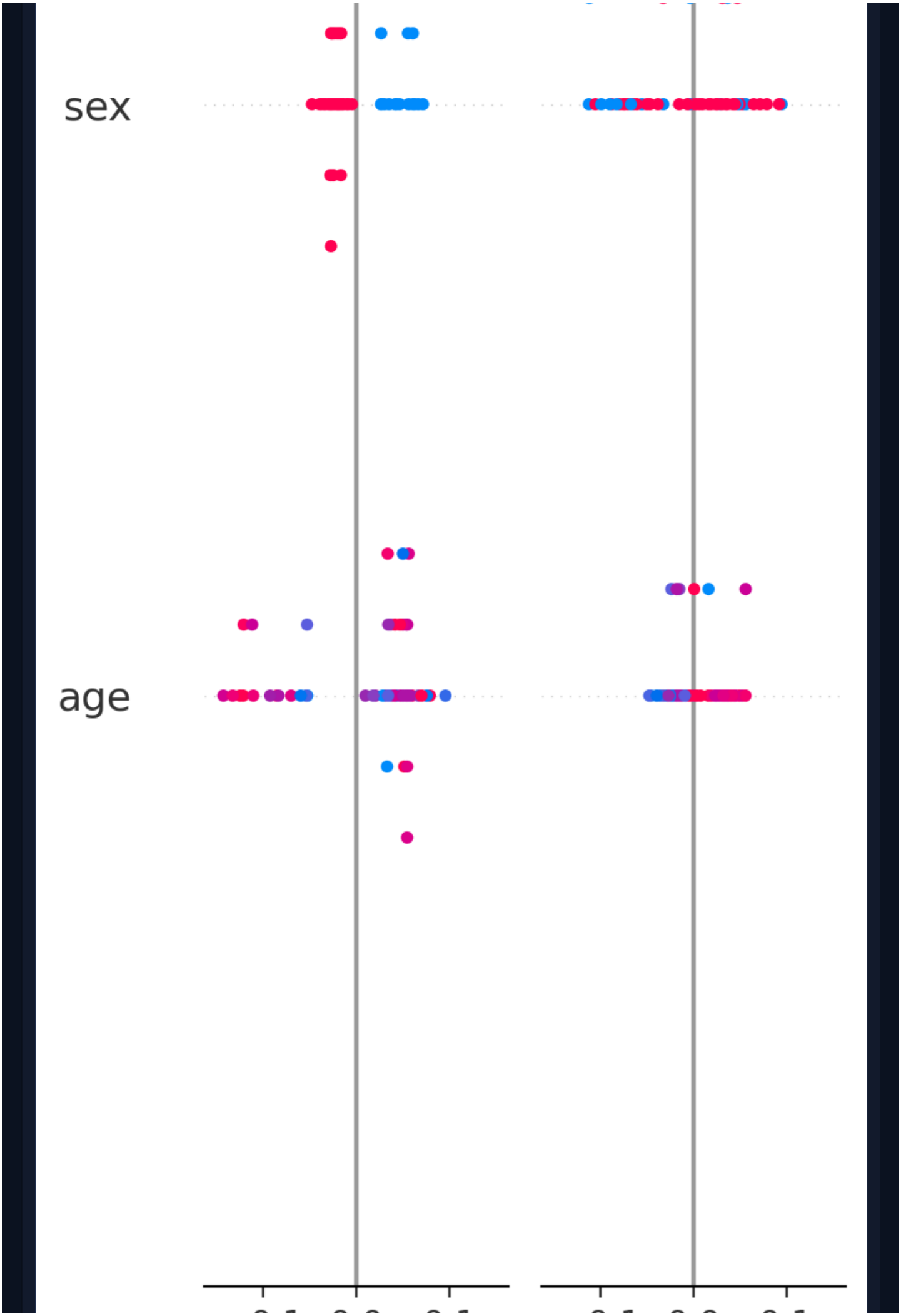
Permutation Importance

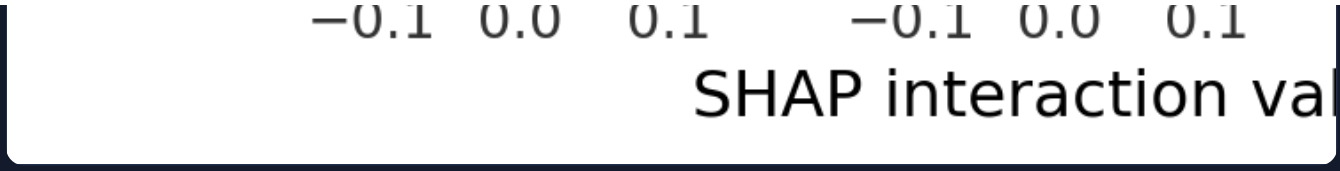


permutation_importance.png

SHAP Summary

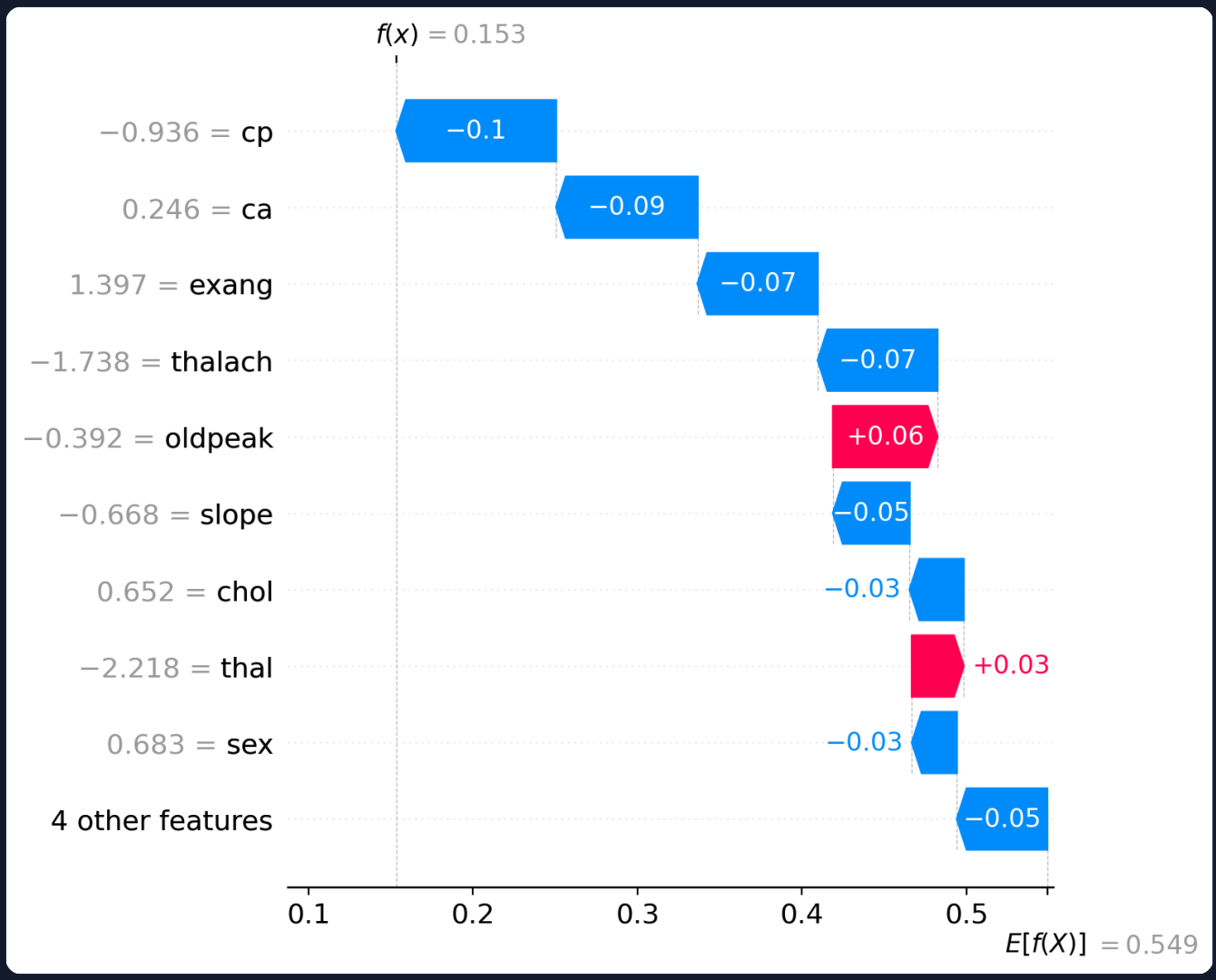



sex
age



shap/shap_summary.png

SHAP Waterfall



shap/shap_waterfall.png

EDA Text Summary (snippet)

```
.join(lines)
```